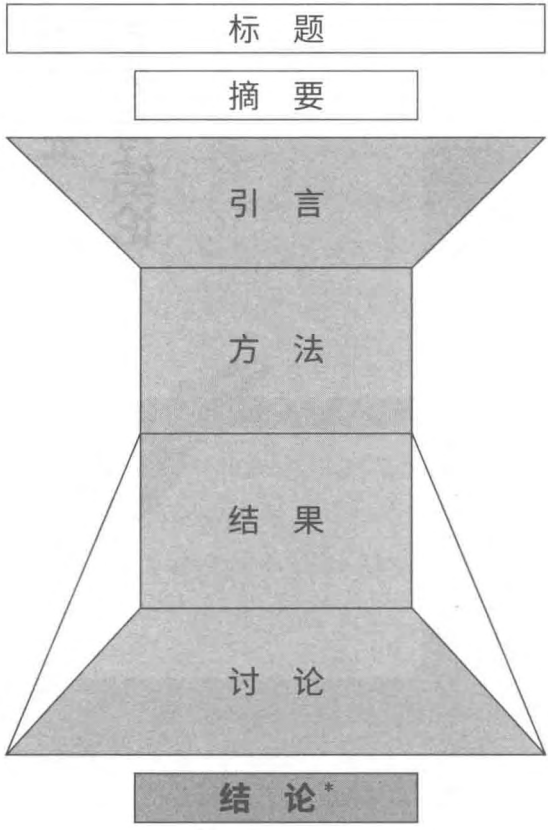
第五单元
怎么写结论





* 有些期刊在这一部分使用结论一词的单数形式，有些使用复数形式，但这似乎并不能反映所得结论的数量。

在一些期刊中，结论部分出现在讨论的最后一两个段落，而在大多数情况下，它是个独立的部分。第三单元的开头针对研究论文的细分方式提出了4种结构选项。下面的表格提供了这4种情况中每一种常见结论部分的详细信息：

<table>
<tr><th>子标题
选项1</th><th>子标题
选项2</th><th>子标题
选项3</th><th>子标题
选项4</th></tr>
<tr><td>Results</td><td>Results</td><td rowspan="2">Results and Discussion</td><td>Results</td></tr>
<tr><td>Discussion</td><td>Discussion</td><td>∅</td></tr>
<tr><td>Conclusion/s</td><td>∅</td><td>Conclusion/s</td><td>Conclusion/s</td></tr>
<tr><td>有一个单独的结论部分，由 1～2 个简短段落组成</td><td>讨论部分最后的 1～2 个段落在形式和内容上与结论部分相似</td><td>有一个独立的结论部分，由 1～2 个简短的段落组成</td><td>可能有一个很长的结论部分，最初在形式和内容上与讨论部分相似，并以 1～2 个段落在形式和内容上与结论部分相似的段落结束</td></tr>
</table>

除了第四个选项外，结论部分单词的总量为100～200个，通常由一个或两个简短的段落组成，有时甚至包含要点列表。然而，结论的长度因不同学科和每个学科的不同期刊而异，因此有必要将目标文章中结论部分的单词平均数作为指导。

结论部分可能包含与前面部分重叠的内容，尤其是摘要（Abstract）、引言（Introduction）和讨论（Discussion）；它和一些

期刊中出现的新模块也有一些重叠，比如梗概（Synopsis）和亮点（Highlights）。然而，结论的功能不同于所有这些，它超越了重复或总结。审稿人、编辑和读者将结论视为一个关键部分，它将传递一个专注于研究结果和影响的明确的关键信息，作者应该完全满足这一期望。

5.1 创建模板

使用第1～4单元中的逆向工程方法列出以下六个结论部分中句子的功能，然后将你的列表与第 5.2 节中的**通用结论模板**进行比较。在前两个例子中，结论部分中句子功能的线索以**粗体**突出显示。在某些情况下，突出显示动词只是为了引起人们对动词时态的注意。

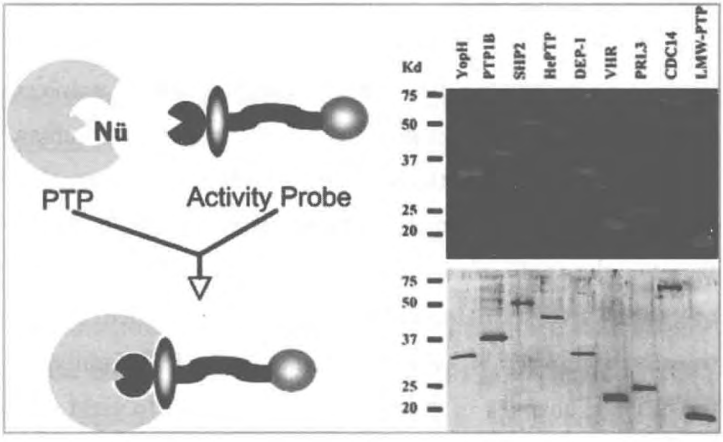
每个结论部分前面都有该论文的**摘要**（Abstract）以及相关的**梗概**（Synopsis）或**亮点**（Highlights），以帮助你了解它们与结论部分的不同之处，并强调摘要不应用作论文结论部分的主要来源材料。

1 Global analysis of protein tyrosine phosphatase activity with ultrasensitive fluorescent probes[t]

Synopsis

Protein tyrosine phosphatases (PTPs) consist of a large

family of enzymes regulating virtually all aspects of cellular processes. This paper describes the development of activity-based chemical probes for global profiling the entire PTP family on the basis of phosphatase activity, which should yield new functional insights into pathways regulated by PTPs and contribute to the discovery of PTPs as novel therapeutic targets.



Abstract

Protein tyrosine phosphatases (PTPs) consist of a large family of enzymes known to play important roles in controlling virtually all aspects of cellular processes. However, assigning functional significance of PTPs in normal physiology and in diseases remains a major challenge in cell signaling. Since the function of a PTP is directly associated with its intrinsic

activity, which is subject to post-translational regulation, new tools are needed to monitor the dynamic activities of PTPs, rather than mere abundance, on a global scale within the physiologically relevant environment of cells. To meet this objective, we report the synthesis and characterization of two rhodamine-conjugated probes that covalently label the active site of the PTPs in an activity-dependent manner, thus providing a direct readout of PTP activity and superior sensitivity, robustness, and quantifiability to previously reported biotinylated probes. We present evidence that the fluorescent probes can be used to identify new PTP markers and targets for potential diagnosis and treatment of human diseases. We also show that the fluorescent probes are capable of monitoring H_2O_2-mediated PTP inactivation, which should facilitate the study of regulated H_2O_2 production as a new tier of control over tyrosine phosphorylation-dependent signal transduction. The ability to profile the entire PTP family on the basis of changes in their activity is expected to yield new functional insights into pathways regulated by PTPs and contribute to the discovery of PTPs as novel therapeutic targets Protein tyrosine phosphatases (PTPs) consist of a large family of enzymes known to play important roles in controlling virtually all aspects of cellular processes. However, assigning functional significance of PTPs in normal physiology and in diseases remains a major challenge in cell signaling. Since the function of a PTP is directly associated with its intrinsic activity, which is subject to post-translational regulation,

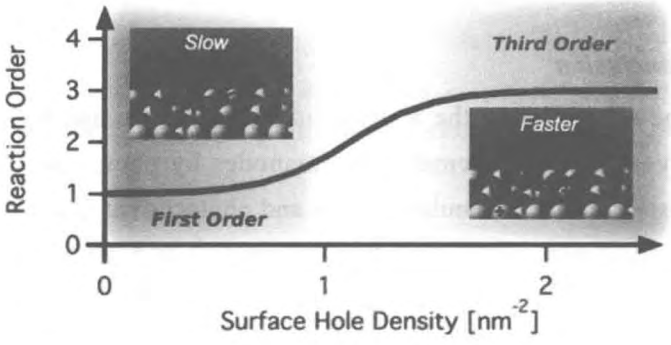
new tools are needed to monitor the dynamic activities of PTPs, rather than mere abundance, on a global scale within the physiologically relevant environment of cells. To meet this objective, we report the synthesis and characterization of two rhodamine-conjugated probes that covalently label the active site of the PTPs in an activity-dependent manner, thus providing a direct readout of PTP activity and superior sensitivity, robustness, and quantifiability to previously reported biotinylated probes. We present evidence that the fluorescent probes can be used to identify new PTP markers and targets for potential diagnosis and treatment of human diseases. We also show that the fluorescent probes are capable of monitoring H_2O_2-mediated PTP inactivation, which should facilitate the study of regulated H_2O_2 production as a new tier of control over tyrosine phosphorylation-dependent signal transduction. The ability to profile the entire PTP family on the basis of changes in their activity is expected to yield new functional insights into pathways regulated by PTPs and contribute to the discovery of PTPs as novel therapeutic targets.

Conclusions

We have synthesized and characterized two fluorescent rhodamine-containing PTP probes that arehighly sensitive for direct in-gel visualization of PTP activity. **Kinetic analyses suggest that** these probes **are** active site directed **and inactivate** a broad range of PTPs in a time- and concentration-

dependent fashion. **Direct in-gel fluorescence scanning indicates that** the fluorescent probes **form** a covalent adduct with the PTPs and the amount of labeling correlates with PTP activity. **As expected**, the fluorescent probes can detect on the order of 100 attomole of rhodamine-labeled PTP, a detection limit **nearly 3 orders of magnitude more sensitive than** that of the biotin-conjugated probes. **Moreover**, the fluorescent probes also **exhibit extremely high** selectivity toward PTPs while remaining inert to other proteins. **Thus**, the rhodamine-containing probes **provide superior** sensitivity, quantifiability, and throughput for activity-based PTP profiling. **Initial proof-of-concept experiments show that** the fluorescent probes **are capable of** monitoring **simultaneously** the activity levels of PTPs at the whole proteome level **and that** the activity profiles of PTPs **are significantly different** among a panel of human cancer cell lines. **This highlights the potential to use** the fluorescent probes to identify new PTP markers and targets for the diagnosis and treatment of human diseases. **Finally, it is shown that** the fluorescent probes **are capable of** monitoring H_2O_2-mediated PTP inactivation, **which should facilitate the study of** regulated H_2O_2 production as a new tier of control over tyrosine phosphorylation-dependent signal transduction. **Further application** of the activity-based fluorescent probes **will accelerate global characterization** of PTPs, **thereby increasing our understanding of** PTPs in cell signaling and in diseases.

2 Rate law analysis of water oxidation on a hematite surface[u]



Abstract

Water oxidation is a key chemical reaction, central to both biological photosynthesis and artificial solar fuel synthesis strategies. Despite recent progress on the structure of the natural catalytic site, and on inorganic catalyst function, determining the mechanistic details of this multiredox reaction remains a significant challenge. We report herein a rate law analysis of the order of water oxidation as a function of surface hole density on a hematite photoanode employing photoinduced absorption spectroscopy. Our study reveals a transition from a slow, first order reaction at low accumulated hole density to a faster, third order mechanism once the surface hole density is sufficient to enable the oxidation of nearest neighbor metal atoms. This study thus provides direct evidence for the multihole catalysis of water oxidation

by hematite, and demonstrates the hole accumulation level required to achieve this, leading to key insights both for reaction mechanism and strategies to enhance function.

Conclusion

The rate order of the water oxidation reaction **has been investigated** on hematite photoanodes by photoinduced absorption of accumulated holes and photocurrent densities recorded simultaneously. A transition from first order to third order in photogenerated holes **has been identified** when a sufficient density of holes is accumulated at the semiconductor-electrolyte interface to oxidize two nearest neighbor surface metal atoms. This transition **was confirmed by** agreement between three different analyses. The third order reaction **is rationalized** by considering the rate-limiting formation of an intermediate, which requires the incorporation of three holes. **It is argued that*** this rate law **is consistent with** reaction mechanisms based on recent observations of intermediates involving oxygen–oxygen bond formation during the oxygen evolution reaction on other metal oxide surfaces.

This study clearly demonstrates the ability of a metal oxide semiconductor surface, such as hematite, to drive a multihole reaction, but **also highlights** the modest catalytic properties of this surface. **Our findings will help to tailor the design of** metal oxide anodes and photoanodes for water oxidation with regards to achieving high localized

hole concentration, and the need for cocatalysts to enhance function. **Further work is currently in progress** with different materials and overlayers to determine their function and abilities to catalyze this complex but fascinating reaction.

* 在这里作者似乎在自我引述，例如 We argue that（我们认为），但这个句子也可能意味着 it is argued by other that...（其他人认为）。

3 Performance of structural concrete with recycled plastic waste as a partial replacement for sand[v]

Abstract

Environmental concerns arising from the over-dredging of sand have led to restrictions on its extraction across India, with direct economic impacts on concrete construction. A suitable environmentally friendly alternative to sand must be found to match the huge demand from the concrete construction industry. At the same time, waste plastic is rarely recycled in India, with as much as 40% left in landfill. The dumping of such materials which degrade at extremely low rates meaning they persist in the environment is a long-term environmental concern.

To tackle both issues, it is proposed to process waste plastic to create a partial replacement for fine sand in a novel mix for structural concrete. In this paper eleven new concrete mixes are evaluated to study five plastic material compositions,

three groups of particle sizes, three different aspect ratios, and two chemical treatments and establish an appropriate choice of material to act as partial replacement for sand.

The results show that replacing 10% sand by volume with recycled plastic is a viable proposition that has the potential to save 820 million tonnes of sand every year. Through suitable mix design the structural performance of concrete with plastic waste can be maintained. This preliminary work was supported through funding from the British Council under the UKIERI (United Kingdom India Educational Research Initiative) programme for the project 'Development of structural concrete with the help of plastic waste as partial replacement for sand'.

Conclusions

This paper has demonstrated the potential for using recycled waste plastic in structural concrete mixes. At a replacement ratio of 10% by volume, this has the potential to save 820 millions tonnes of sand every year from being used in concrete mixes [1]. This is equivalent to approximately 5% of total global annual sand consumption. A further benefit is to add value to waste plastic, helping to reduce the volumes sent to landfill in some countries. A reduction in sand demand from the construction industry would further support efforts to limit the effects of sand dredging in countries such as India and China, where significant sand volumes are extracted every year.

It is generally seen that substituting plastic into a concrete

mix causes a decrease in compressive and tensile strength due to the poor bond between the plastic and surrounding matrix. Since failure in concrete propagates in tension, the poor bond around plastic particles leads to a reduced compressive and tensile strength. The use of a graded PET plastic matched to the size of the sand particles it replaces, and at a replacement of 10% by volume, gave the most promising overall performance. This material is furthermore cost effective to produce and comes widely available as a waste material in many markets. This paper has shown that simply shredding a PET material is sufficient processing to provide a viable alternative to sand.

Testing different forms of plastic has demonstrated that the most efficient plastic aggregate used in a concrete mix should have a rough surface, be irregular in shape, and be sufficiently small so as to not create a significant failure surface, but also be graded similar to the sand it replaces. The results indicate that through appropriate mix design reductions in strength can be minimised to acceptable levels.

4 The oxidative corrosion of carbide inclusions at the surface of uranium metal during exposure to water vapour[d]

Abstract

The reaction between uranium and water vapour has been

well investigated, however discrepancies exist between the described kinetic laws, pressure dependence of the reaction rate constant and activation energies. Here this problem is looked at by examining the influence of impurities in the form of carbide inclusions on the reaction. Samples of uranium containing 600 ppm carbon were analysed during and after exposure to water vapour at 19 mbar pressure, in an environmental scanning electron microscope (ESEM) system. After water exposure, samples were analysed using secondary ion mass spectrometry (SIMS), focused ion beam (FIB) imaging and sectioning and transmission electron microscopy (TEM) with X-ray diffraction (micro-XRD). The results of the current study indicate that carbide particles on the surface of uranium readily react with water vapour to form voluminous $UO_3 \cdot xH_2O$ growths at rates significantly faster than that of the metal. The observation may also have implications for previous experimental studies of uranium–water interactions, where the presence of differing levels of undetected carbide may partly account for the discrepancies observed between datasets.

Highlights

High resolution imagery (FIB, SEM and SIMS) of carbide inclusions in uranium metal. ► Real time images following the reaction of the carbide inclusions with water vapour. ► Shown preferential consumption of carbide over that of the bulk metal. ► Quantity of impurities in the metal therefore

seriously influence reaction rate. ▶ Metal purity must be considered when storing uranium in air or moist conditions.

Conclusions

Examination of cast α-uranium surfaces after exposure to water vapour in an ESEM instrument at 19 mbar and 20 °C indicated surface corrosion had occurred, with secondary growths determined to be $UO_3 \cdot H_2O$ (metaschoepite) forming at carbide inclusions across the surface. Over the period of a week the growths were observed to increase in size, consuming only the carbide particles and not the surrounding metal. In some cases complete decomposition of the surface carbides was observed.

From the results of the current study it is apparent that the carbide particles reacted more readily with the water vapour than the metal. Resultantly it is suggested that disparities between previous studies of the uranium–water reaction may be attributable to differential purities of uranium metal used by different research groups, with resultantly different populations of carbide particles.

5 Measurement and modeling of the phase behavior of the (carbon dioxide + water) mixture at temperatures from 298.15 K to 448.15 K[w]

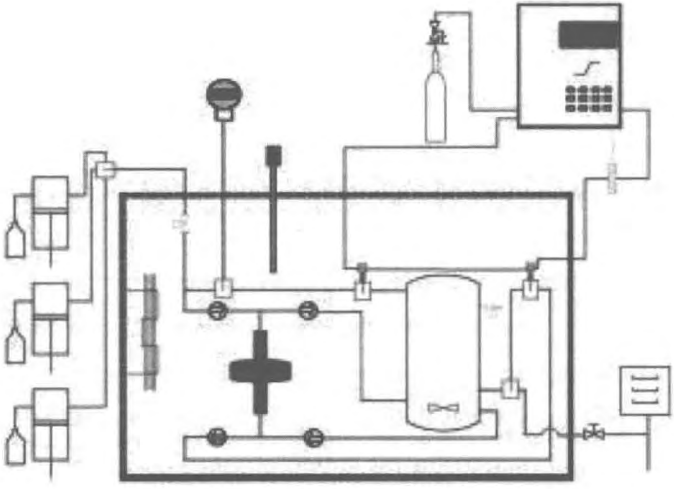
Abstract

An analytical apparatus has been designed to study the phase behavior of fluid mixtures of relevance to CO_2 -enhanced oil recovery and carbon dioxide storage in deep aquifers or depleted oil fields. The fluid phases are circulated by means of a dual-channel magnetically-coupled pump and aliquots may be withdrawn from the re-circulation loops, by means of high-pressure sampling valves, for analysis by gas chromatography. The high-pressure cell is fitted with a special probe that may be rotated in order to draw liquid into the recirculation loop from different heights within the cell, thereby permitting the study of three-phase vapor–liquid–liquid equilibria. The working temperature range of the apparatus is from (298 to 448) K and the maximum working pressure is 50 MPa.

In this work, measurements have been made on the binary system (CO_2 + H_2O) at temperatures from (298.15 to 448.15) K and pressure from (1.5 to 18.0) MPa, and the results are compared with the available literature data. Vapor–liquid–liquid and liquid–liquid equilibrium points were also measured at T = 298.15 K. Standard uncertainties were 0.04 K for temperature, 0.04% of reading for pressure, and typically 3×10^{-4} and 8×10^{-4} for the mole fractions in

liquid and vapor phases respectively. The results have been correlated by means of an asymmetric approach based on the Peng–Robinson equation of state, for the vapor phase, and an extended form of Henry's law incorporating the NRTL solution model, for the aqueous liquid-phase. The ability of the Krichevsky– Kasarnovsky (KK) approach to correlate the data has also been evaluated.

Graphical abstract



Highlights

► An analytical apparatus has been designed and commissioned for high T and p phase behavior measurements.

► Measurements have been made on CO_2 + H_2O at temperatures from (298.15 to 448.15) K and pressures from (1.5 to 18.0) MPa. ► The measured data have been correlated by means of an asymmetric approach. ► The Krichevsky–Kasarnovsky approach and Duan's correlation model are also evaluated.

Conclusions

The phase behavior of (CO_2 + H_2O) was measured at temperature from 298.15 K to 448.15 K, and at pressures up to 18 MPa. The experimental results are compared comprehensively with literature data and found to agree with those literature sources identified in earlier reviews as being of the highest reliability. Our results fill key gaps in terms of accurate and high-quality data at high temperature and pressure, and pave the way for measurements on (CO_2 + brine) systems. The results are modeled accurately with a γ–φ approach incorporating the Peng–Robinson EoS with the classical mixing rules for the vapor phase, and an extended form of Henry's law with the NRTL solution model and a Poynting correction for the liquid phase. The Krichevsky–Kasarnovsky (KK) (simplified γ–φ approach) was also studied as well as the empirical correlation of Duan *et al.* Both were found to have significant deficiencies for this system.

6 Computer-controlled stimulation for functional magnetic resonance imaging studies of the neonatal olfactory system[x]

Abstract

Aim: Olfactory sensation is highly functional early in human neonatal life, with studies suggesting that odours can influence behaviour and infant–mother bonding. Due to its good spatial properties, blood oxygen level–dependent (BOLD) contrast functional magnetic resonance imaging (fMRI) has the potential to rapidly advance our understanding of the neural activity which underlies the development of olfactory perception in this key period. We aimed to design an 'olfactometer' specifically for use with neonatal subjects for fMRI studies of odour perception.

Methods: We describe a fully automated and pro-grammable, fMRI compatible system capable of presenting odorant liquids. To prevent contamination of the system and minimize between-subject infective risk, the majority of the olfactometer is constructed from single-use, readily available clinical equipment. The system was used to present the odour of infant formula milk in a validation group of seven neonatal subjects at term equivalent postmenstrual age (median age 40 weeks).

Results: A safe, reliable and reproducible pattern of stimulation was delivered leading to well-localized positive

BOLD functional responses in the piriform cortex, amygdala, thalamus, insular cortex and cerebellum.

Conclusions: The described system is therefore suitable for detailed studies of the ontology of olfactory sensation and perception during early human brain development.

Key notes

- Ofaction is important in early life, but the anatomical substrates of the underlying neural activity are poorly understood.
- We describe a fully automated and safe system for fMRI studies of olfaction in neonatal subjects.
- Functional activation can be identified with fMRI in the primary olfactory areas in the neonatal brain.

Conclusions

We have designed, constructed and implemented a fully automated and programmable olfactometer specifically for fMRI studies of neonatal subjects. The described system is safe, has been designed to minimize infective risks and produces a reproducible but flexible pattern of olfactory stimulation. The system was found to induce a well-localized pattern of positive BOLD functional activity in the primary olfactory areas of seven neonatal subjects at term equivalent PMA. These data suggest that fMRI now offers the opportunity to investigate the ontogeny of olfaction in

the human newborn with millimetre-scale precision, and to address specific hypotheses concerning the development of mature olfactory responses.

5.2　测试和调整模板

在本单元上面列出的结论部分中，你找到了多少关于句子、句子的组成部分或语言的例子呢？这些例子可以表达或回顾下面通用结论模板中的组成部分。

通用结论模板	
1	论文里包含什么
2	论文/研究取得了什么成就
3	相关背景信息
4	研究的空白/目标/需求
5	方法/途径
6	带有评价性评论的关键结果
7	结果的含义
8	潜在或实际的局限性
9	潜在或实际的应用
10	该研究如何提高知识
11	未来可能的研究方向

将组件1～11映射到你的目标论文的结论部分，以查看它们与本单元中的六个结论部分的相似或不同之处。你的分析应

考虑各组件通常出现的顺序，以及每个组件在结论部分所占的比例。

5.3 有用的词汇和短语

结论的所有组成部分的语言建议已出现在前面的单元中。例如：

1.	论文里包含什么	参看第一和第二单元
2.	论文/研究取得了什么成就	参看第四单元
3.	相关背景信息	参看所有单元
4.	研究的空白/目标/需求	参看第一单元
5.	方法/途径	参看第二和第三单元
6.	带有评价性评论的关键结果	参看第二和第三单元
7.	结果的含义	参看第三单元
8.	潜在或实际的局限性	参看第二和第三单元
9.	潜在或实际的应用	参看第四单元
10.	该研究如何提高知识	参看第四单元
11.	未来可能的研究方向	参看第四单元

5.4 语言和写作技巧

5.4.1 结论的动词时态

在结论部分，一般现在时和现在完成时常用于表达论文

中的内容（what **is** in the paper）、结果所表示的内容（what **is** implied by the results）以及研究中取得的成果（what **has been** achieved in the study）：

- **We have synthesized and characterized** two fluorescent rhodamine-containing PTP probes…
- **It is argued that** this rate law is consistent with reaction mechanisms based on recent observations…
- This long-standing problem **is solved** by combining micronsized and nano-sized toughenersin a synergistic manner.
- The rate order of the water oxidation reaction **has been investigated on** hematite photoanodes by…
- In the present paper, some shortcomings of the concept of cross-section classification **have been highlighted**, and, as an alternative treatment, the continuous strength method **has been introduced**.
- The level of enhancement in resistance offered by the continuous strength method over conventional design methods for steel structures **has been found to be** approximately…
- Finally, **it is shown that** the fluorescent probes are capable of monitoring…
- Our results **fill** key gaps in terms of accurate and high-quality data at high temperature and pressure, and **pave the way for** measurements on (CO_2 + brine) systems.
- We **have designed, constructed and implemented** a fully

automated and programmable olfactometer specifically for fMRI studies of neonatal subjects.

- The described system **is safe, has been designed** to minimize infective risks and **produces** a reproducible but flexible pattern of olfactory stimulation.

一般过去时态不太常见，但有时也用于重复方法的关键方面或关键的结果：

- The phase behavior of (CO_2 + H_2O) **was measured** at temperature from 298.15 K to 448.15 K,...
- Damage accumulation **evolved** slowly at lower strains, but **continued** at a rapid rate immediately prior to fracture, which is a typical ductile fracture characterized by dimples.
- In some cases complete decomposition of the surface carbides **was observed**.
- The system **was found to** induce a well-localized pattern of...

5.4.2 拥有你的贡献

在结论中指明你的贡献并不简单，它需要你明确无误地表达。作者有时会错误地假设哪些句子能明显地呈现他们自己的贡献。这个错误的假设有两个原因：(1)相信读者在阅读结论之前已经阅读了论文的其余部分；(2)作者自己对项目及其结论的熟悉程度。

大多数读者在阅读结论之前不会阅读整篇论文。就像阅读谋杀之谜的读者可能会直接跳到最后一页找出谁犯了谋杀罪一样，研究论文的读者可能会在某个时候跳到结论部分去找出文本的去向。在阅读论文的其余部分之前，直接进入结论可能有助于读者更有效地处理中间信息，或者它本身可能就是一个目的，即读者不打算阅读中间部分。一些读者直接从摘要，或者甚至从标题跳到结论。然而，还要注意，即使读者阅读了整篇论文直到结论，也不能假设他们完全准确地保留了前面部分中的所有信息。

作者自己对内容的熟悉程度也是一个风险因素。在研究项目期间，这种熟悉程度随着每天都与同事分享而增加，并延伸到写作阶段，使作者对潜在的歧义和更加明确的表达需要视而不见。如果你和你的同事理解你所写的内容，便似乎没有必要将事情说得更清楚——你甚至可能会觉得是屈尊俯就的。但是，带有非人物语法主语和动词的一般现在时的句子并没有告诉读者这些结论事实上是这项研究所得到的，例如下面的这个句子：

Kinetic analyses suggest that these probes are active site directed and inactivate a broad range of PTPs in a time-and concentration-dependent fashion. **Direct in-gel fluorescence scanning indicates** that the fluorescent probes form a covalent adduct with the PTPs.

没有告诉读者这些实际上是这项研究的结论。这些句子可以很容易地呈现**相关背景信息**(relevant background information)或**研究的空白/目标/需求**(the gap/aim/need for the study),这两种情况有时会出现在结论中(参见第5.2节)。读者不会进行法医式的阅读,他们的阅读速度很快,而在这个关键点上快速阅读这些句子有遗漏其归属的风险。

在上面的结论部分,作者可以通过使用以下一项或多项措施来避免歧义:

- 一般过去时或现在完成时
- 主动而不是被动
- 人物语法主语
- 明确的句子开头或明确提到论文或研究的语言

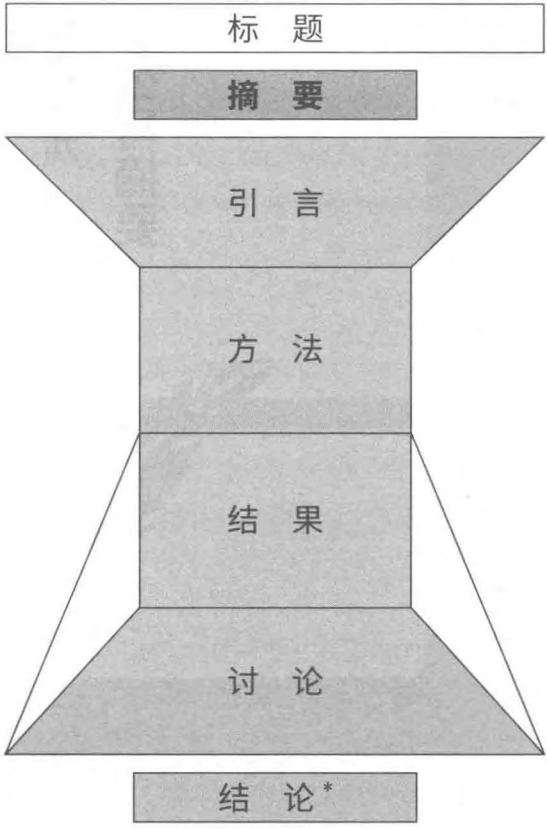
下面是这四项应用的例子:

- **We have synthesized and characterized** two fluorescent rhodamine-containing PTP probes that are highly sensitive for direct in-gel visualization of PTP activity.
- Examination of cast α-uranium surfaces after exposure to water vapour in an ESEM instrument at 19 mbar and 20 °C **indicated that** surface corrosion had occurred,…
- This transition **was confirmed by** agreement between three different analyses.

- **Finally, it is shown here that** the fluorescent probes are capable of monitoring H_2O_2 -mediated PTP inactivation,…
- **This study clearly demonstrates** the ability of a metal oxide semiconductor surface, such as hematite, to drive a multihole reaction,…
- **In the present paper**, some shortcomings of the concept of cross-section classification **have been highlighted**, and, as an alternative treatment, the continuous strength method **has been introduced**.
- **From the results of the current study** it is apparent that the carbide particles reacted more readily with the water vapour than the metal.

第六单元
怎么写摘要





* 有些期刊在这一部分使用结论一词的单数形式，有些使用复数形式，但这似乎并不能反映所得结论的数量。

摘要是一份意义重大的文件：比起整篇论文，有更多的人阅读摘要，或简要地浏览它。好的摘要可以提高研究的可见度，而一个糟糕的摘要可能会导致研究被忽视。

近年来，摘要的写作发生了巨大变化，并且随着研究交流的发展和变化不断演变。互联网阅读意味着摘要不再是作为一个为研究论文设计的序曲或附件，它通常是被单独阅读的，并且必须作为一个单独的、独立的文本运行，阅读它的读者会出于各种原因对其进行访问。有些读者只是想知道特定研究领域当前正在发生的事情，而可能对技术细节不感兴趣；另一些可能是该领域的专家，他们想了解细节，但只对与他们自己的项目直接相关的研究感兴趣。读者可能只是出于一个特定的原因去阅读摘要，例如，他们可能只想检查他们自己当前或计划的项目是否被另一个研究小组覆盖。另一种可能性是读者可能只想知道该研究的应用，例如，在医学上，治疗决定有时是仅基于摘要来做出的。

每年有超过三百万篇研究论文发表，当读者滚动浏览标题和摘要时，创造了一种研究在网上争夺注意力的环境。然而，写作指南有时不一致或令人困惑，例如，摘要应该“在简明扼要的同时包括所有相关的信息”这样的建议。在其他情况下，建议又过于模糊，一个写得“清楚”和“连贯”的建议需要提供如何这样做的实际细节。作者总是理解他们所写的内容，因此认为它是清晰的，但那些连续快速浏览几篇摘要的读者需要非常高的清晰度和连贯性。

6.1 摘要指南

清晰度和连贯性

清晰度和连贯性可以通过以下方式提高：

- **明确地指明你的贡献**。你对这项研究的熟悉程度使得你更难看到摘要中潜在的歧义，而这在指明你自己的贡献方面尤其重要。当使用一般现在时态的时候，信息似乎是众所周知的，而实际上它可能是对你的结果的描述。像**理论建模表明**（theoretical modelling suggests that...）这样的短语可能指已确立的知识，也可能指研究论文本身的内容和贡献。为了避免歧义，请考虑：
  1. 用一般过去时或现在完成时："Analysis shows concordance for 74% of mutation calls."（不清楚是谁进行了分析），相比 "Analysis showed concordance for 74% of mutation calls."（是作者进行了分析）。
  2. 使用主动语态（we）而不是被动语态。
  3. 在句子的开头使用像It is known that/It has been previously demonstrated that 或 In this study/Here的短语。

摘要中严格的词数限制意味着你可能会试图省略这些短

语，但**你的目的不是让读者能够理解你的摘要，而是让读者不可能理解不了你的摘要和识别你的贡献**。

- **避免指代词的歧义**。指代词像**it**、**which** 或**this** 可能对你而言是清楚易懂的，但对读者则不然。如果可能存在歧义，请在**this**之类的词后面加上适当的名词或重复名词/短语。正如你将在摘要4（第6.2.1节）中看到的，为了避免歧义，短语 the hybrid method 在一个140个单词的摘要中重复了至少4次。
- **使用一致的语言**。如果某个东西被描述为一种**途径**（approach），它不应该突然成为一个**方案**（scheme）。类似地，一个**方案**（scheme）不应该突然成为一个**框架**（framework），一个**模型**（model）不应该突然成为一种**方法**（method），一个**工具**（tool）也不应该突然成为一个**设备**（device）。这样会模糊你提供的内容的身份，也使摘要难以理解。
- **使用第一单元中描述的句子开头的技巧**去创建一个强有力和利于读者的叙述。
- **保持句子长度简短**。少于20个单词的句子在第一次阅读时能被90%的读者理解，那些超过40个单词的句子在第一次阅读时只能被10%的读者理解。本单元第6.2小节中的18个摘要的平均句子长度为22个单词。

摘要应该有多长?

将目标期刊摘要中的字数进行平均会给你一个粗略的主意。大多数摘要都在80～250字之间，并被写成一个单独的段落。

字数限制不是目标。因为摘要是一个如此意义重大的文本，你会忍不住使它超载信息。但是，如果无视相关性或叙述连贯性而将空间填充到最大允许的字数，会使关键的研究信息变得模糊不清。摘要不一定是整篇论文的概述。尽管很想将研究的所有成就都包含在摘要中，但这么做会分散影响力，读者实际上可能无法发现核心的价值或贡献。优先考虑核心成就或贡献，并明确地向读者发出信号。

我应该什么时候写摘要?

摘要通常是在论文的其他部分都完成以后才写的。这么做有许多的原因：

1. 如果摘要写得太早，就有可能与论文的内容不一致或包含了不在论文的最终版本中的信息。论文在投稿之前经常需要更改或更新，因此如果摘要早已经写好，应进行审查，有必要的话，在投稿前修改或重新调整摘要重点。

2. 摘要的内容、风格和长度取决于你计划在哪里投稿，并且投稿决定可能会在写作过程的后期做出。在投稿前最后一分钟剪切和粘贴以重新调整摘要，使其符合新的投稿指南，这么做可能会有失去连贯性的风险。

3. 摘要源于这篇论文，而不是其他不相关的内容，因此它

不需要“创建”；它需要仔细选择相关和适当的内容，以及易于阅 读的强有力的指导性叙述。

我应该用什么样的语言？

摘要的语言应该尊重全球读者、跨学科读者和快速读者的需求——换句话说，它应该是简单明了的。摘要中使用的非技术词汇在所有学科中都是相似的，这反映了学术写作对清晰度和可获取性的需求。请参照本单元第6.4小节的例子。

一些读者可能不熟悉你想在摘要中使用的特定技术用语或者首字母缩拼词，这将限制他们对摘要内容的获取，并可能影响他们继续阅读的意愿。编辑不喜欢充满简称的摘要，故而如果因为摘要中不断重复完整词语可能影响字数限制而确实需要首字母缩略词，那么（一次性）指定简称代表什么可能是个好主意。

下面是一些摘要应用的例子：

- We have developed and implemented a robust and practical scheme for anisotropic 3D acoustic full-waveform inversion (FWI)…
- This study aims to quantify and compare the accuracy of traditional radiostereometric analysis (RSA)…
- While brain–computer interfaces (BCIs) can provide communication to people who are locked-in, they suffer from a very low information transfer rate. Further,

using a BCI requires a concentration effort and using it continuously can be tiring. The brain controlled wheelchair (BCW) described in this paper aims at providing mobility to BCI users…

6.2 摘要的类型

摘要有四种主要类型：

1.简单的/一般的摘要

2.有结构的摘要，具有如**背景/目的/方法/结果**（**Background/Aims/Method/Results**）等的标题

3.与**重要性说明**（**a Significance Statement**）或**亮点**（**Highlights**）的列表在一起的摘要

4. 图形摘要

6.2.1 简单的/一般的摘要

利用前面单元中演示的逆向工程技术，根据以下十个简单的/一般的摘要中的句子功能创建模板。在这些摘要中，模板的标识符以**粗体**显示。完成后，在第6.2.2节的结构化摘要中为模板标识符画线或做标记，以支持你的模板，然后根据第6.3节中的通用模板检查和调整它。

1 A sub-mW fully-integrated pulse oximeter front-end[y]

Abstract

This paper presents the implementation of the first fully integrated pulse oximeter front-end with a power consumption lower than 1 mW. **This is enabled by** system- and block-level noise optimisation, **also detailed in the manuscript**. The proposed design features an analogue feedback loop **that enables fast and accurate** regulation of the detected photocurrent level and a serial-to-parallel interface **allowing for** extensive programmability of several operation parameters. The front-end **was fabricated** in the AMS 0.35 μm technology and occupies **an occupies an area of 1.35 mm^2. Extensive measured results,** both electrical and physiological from human subjects **are reported, demonstrating** an estimated SNR of 39 dB and **ability to** detect 2% changes in SpO2, **similar to** commercial pulse oximeters. **This is despite** the constrained power consumption which amounts to 0.31 mW for the LEDs and 0.53 mW for the rest of the front-end from a 3.3 V supply. Statistical **results** from 20 chips **verify good matching** across the Red and Infrared channels of the front-end **and the accurate operation of the proposed** analogue feedback loop.

2 Anisotropic 3D full-waveform inversion[z]

Abstract

We have developed and implemented a robust and practical

scheme for anisotropic 3D acoustic full-waveform inversion (FWI). **We demonstrate this scheme** on a field data set, **applying it** to a 4C ocean-bottom survey over the Tommeliten Alpha field in the North Sea. This shallow-water data set **provides** good azimuthal coverage to offsets of 7 km, with reduced coverage to a maximum offset of about 11km. The reservoir **lies** at the crest of a high-velocity antiformal chalk section, overlain by about 3000 m of clastics within which a low-velocity gas cloud **produces** a seismic obscured area. **We inverted only** the hydrophone data, **and we retained** free-surface multiples and ghosts within the field data. **We invert** in six narrow frequency bands, in the range 3 to 6.5 Hz. At each iteration, **we selected only** a subset of sources, **using** a different subset at each iteration; **this strategy is more efficient than** inverting all the data every iteration. **Our starting** velocity **model was obtained using** standard PSDM model building including anisotropic reflection tomography, and contained epsilon **values as high as** 20%. **The final** FWI velocity **model shows** a network of shallow high-velocity channels that **match similar features** in the reflection data. Deeper in the section, the FWI velocity **model reveals** a sharper and more intense low-velocity region associated with the gas cloud in which low-velocity fingers **match** the location of gas-filled faults visible in the reflection data. **The resulting** velocity **model provides a better match** to well logs, **and better** flattens common image gathers, than does the starting model. Reversetime migration, **using the** FWI velocity **model, provides significant**

uplift to the migrated image, **simplifying** the planform of the reservoir section at depth. The workflows, inversion strategy, and algorithms **that we have used have broad application** to invert a **wide range of analogous data sets**.

3 Mobilizing salt: magma-salt interactions[aa]

Abstract

Salt sequences **form** an integral part of **many** sedimentary basins **worldwide. Many** of these basins have experienced igneous activity either syn- or post-deposition of the salt sequences. **Despite this, little work has so far been undertaken to understand** magma-salt interactions within the subsurface, and how aspects such as salt halokinesis may be influenced by igneous activity. **Within this paper, we detail the first direct description of** relationships and textures that are developed during intrusive igneous-salt interaction. **We show that** salt composition **appears to play a dominant role** in controlling where igneous intrusions invade laterally through salt sequences in a sedimentary basin. **In particular, we illustrate that** hydrous salts, such as carnallite, **act** as preferential horizons for lateral magma intrusion. This lithological control **appears primarily related to** the heating and subsequent dehydration reaction of carnallite, which causes the carnallite to behave as viscous fluidal horizons, resulting in the non-brittle emplacement

of magma, and spectacular peperitic salt-magma mingling textures. **We suggest that** heating and transformation of carnallite and other hydrous salts into viscous fluidal horizons during igneous intrusion within a regional salt sequence **may act as a possible trigger for** contemporaneous halokinesis, by creating fluid-like viscous detachment layers. Over longer time scales, however, a solidified rigid boxwork of dikes and sills **may create** zones of increased mechanical strength that will locally inhibit further salt flow.

4 A hybrid displacement estimation method for ultrasonic elasticity imaging[bb]

Abstract

Axial displacement estimation **is fundamental to many** freehand quasistatic ultrasonic strain imaging systems. **In this paper, we present a novel estimation method that combines the strengths of** quality-guided tracking, multi-level correlation, and phase-zero search to **achieve high levels of accuracy and robustness. The paper includes a full description of** the hybrid method, in vivo examples to illustrate the method's clinical relevance, and finite element simulations to assess its accuracy. Quantitative and qualitative **comparisons are made with leading** single- and multi-level **alternatives**. In the in vivo examples, the hybrid method **produces fewer** obvious peak-hopping **errors**, and in simulation, the hybrid